

Rustom Murgai Dubash

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Education

University of Pennsylvania, MSE Computer and Information Science (*Submatriculation*)

Haverford College, BSc Computer Science and Economics

Graduation: May 2027

- GPA: 3.93/4.0 | Varsity Cricket Team

Publications

Rustom M. Dubash and Sorelle A. Friedler. 2026. **Self-Promotion in LLM Recommendations**. In *18th ACM Web Science Conference (WebSci '26)*, May 26–29, 2026, Braunschweig, Germany. ACM.

Experience

AI Solutions Engineer Intern, ContinuServe, Marietta, GA

June 2026 – Present

- Built a **Financial Forecasting Agent** that ingests financial datasets, using **Claude API** through a custom **MCP** server allowing the agent to query, join and aggregate the data, producing revenue forecasts and 13-week cashflow reports.
- Integrated **NVIDIA KumoRFM** and **Meta Prophet** to generate forecasts accurate to **>97% mean prediction accuracy** and deployed the agent to the **ContinuFlow AI Forecasting Hub** with a **ReactJS** UI to serve MUC and MME clients.

Computational Biology Researcher, Wan Lab, University of Nebraska Medical Center

June 2026 – Present

- Studying automated **cell-type annotation** for **single-cell RNA-seq** data under Prof. Shibiao Wan, working to improve upon established **reference-based** and **marker-based** annotation methods such including **SingleR**, **Garnett**, and **scDiagnostics**.

Teaching Assistant: Introduction to Computer Science & Data Science, Haverford College

Sept 2025 – Jan 2026

- Held weekly office hours to support **80+ students'** learning in Python programming, **algorithms**, and **data analysis**.

Applied Data Institute Scholar '25, Equitech Futures

Sept 2025 – Dec 2025

- Selected as one of **16 fellows** for a **10-week**, fully funded (**\$9,500**) global fellowship in leadership, data science, and AI.
- Created **KPIs** for African water utility managers, impacting **5M+ citizens** in partnership with **Athena Infonomics** and developed dashboards using an **AI-driven RAG system** using **ChromaDB**, **text embeddings**, and chatbot interface.

Machine Learning Researcher, Mathieson Lab, Haverford College

May 2025 - Aug 2025

- Designed **Transformer sequence models** in PyTorch, analysed **attention mechanisms**, context length, and inductive biases on **next-token prediction tasks**, and used **multiprocessing** and **GPU acceleration** to enable large-scale sequence modeling.
- Implemented long-context alternatives to **quadratic self-attention**, including hybrid convolutional and **sparse-attention architectures** such as HyenaDNA, increasing context length by **12×** baseline Transformer without compromising accuracy.

Research Assistant, Dept. of Economics, National University of Singapore

June 2024 - Aug 2024

- Built a web-scraper to extract **10K+ financial records** for a study on political corruption in Indian State Legislatures.

Selected Projects

Green Bone Protocol (Based on Jade City by Fonda Lee), Python, Anthropic API, Agent Orchestration

- Designed a **tiered multi-agent framework** that routes tasks across models by capability (inspired by Jade City by Fonda Lee) with a dedicated **validation tier** for output checking, and ran a **token-cost study** quantifying the accuracy-vs-cost tradeoff of delegating work down the model hierarchy.

The Outreach Desk - Personalized Cold-Outreach Pipeline, Python, Claude API, Gmail API

- Built a configurable **four-stage pipeline** that scrapes target context, writes a personalized message per target with a single **Claude** call, and resolves contact emails, with **YAML-driven** target types that swap tone and research focus.

Residual-Boosted Transformer Ensembles (Compute-Accuracy Tradeoffs), TensorFlow/Keras, Ensembling

- Implemented **residual boosting** over **ensemble logits** by fitting **probability residuals** with weak Transformers, achieving **93% accuracy** on Yelp Polarity benchmark dataset with a 10M parameter budget (SOTA: 98% accuracy, 355M parameters).

Morphology-Focused Learning for Histopathology, TensorFlow, Grad-CAM, PCA/KMeans, Robustness testing

- Trained a **morphology-focused model** using color jitter and grayscale augmentation, achieving identical performance on RGB vs. grayscale images (**acc. 0.84**), indicating reduced reliance on color and stronger dependence on structural features.

Skills

Programming: Python, Java, JavaScript, HTML, CSS, ReactJS, NodeJS, SQL

Frameworks/Libraries: PyTorch, TensorFlow, pandas, numpy, scikit-learn, Flask, Streamlit, Pygame, Biopython, ChromaDB

ML / AI: Multi-agent systems, Model Context Protocol (MCP), RAG, vector databases, KumoRFM, Prophet, FinBERT, Grad-CAM

Tools/Platforms: Git, VSCode, Render, LaTeX, Excel, Microsoft 365, STATA, BLAST, Phobius, Interpro, UniProt